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A Data-Driven Convergence-Finance Analysis of Monetary-Policy Shocks and Cryptocurrency Market Revaluation

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데이터 기반의 융합 금융 분석: 통화정책 충격과 암호화폐 시장의 재평가

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Abstract This study develops a data-driven convergence-finance framework for examining how major U.S. monetary-policy movements are associated with cryptocurrency market revaluation. Using monthly data on total cryptocurrency market capitalisation (TMC), Bitcoin market capitalisation (BTC), and the U.S. policy-rate series, the analysis separates short-run volatility disturbance from medium-run valuation adjustment. Policy-rate changes are classified into easing and tightening episodes, and post-shock responses are evaluated using event-window statistics, local projections, and ridge-regularised distributed-lag diagnostics. The results indicate that the main post-shock signal is observed in valuation levels rather than volatility. Easing episodes are followed by positive medium-run revaluation in both TMC and BTC, while tightening responses are weaker and less precisely estimated. The comparison between TMC and BTC suggests that market-wide capitalisation provides a useful benchmark for monitoring digital-asset adjustment beyond Bitcoin-specific movements.

Key Words : Cryptocurrency market, Bitcoin, Monetary policy, Market capitalisation, Convergence finance

요 약 본 연구는 주요 미국 통화정책 변화가 암호화폐 시장의 가치 재평가의 연관성을 찾기 위해 데이터 기반 융합 금융 분석 체계를 제시한다. 특히, 연구는 총 암호화폐 시가총액(TMC)과 비트코인 시가총액(BTC), 미국의 정책금리 월별 자료를 이용하여, 단기 변동성 충격과 중기 가치 조정 효과를 구분해 분석한다. 정책의 금리 변동은 완화, 긴축 사건으로 분류되며, 충격 이후 반응은 사건 구간 통계 및 국소 예측과 능형 정규 분포 시차 진단을 통해 평가한다. 분석 결과, 충격 후 주반응은 변동성보다 가치평가 수준에서 관찰된다. TMC와 BTC는 완화 국면에서 중기 구간의 긍정적 가치 재평가가 나타난다. 반면에, 긴축 국면은 추정 정밀도가 낮게 관찰된다. 따라서 두 시가총액의 비교는 시장 전체 시가총액이 비트코인보다 더 디지털자산 시장의 조정 양상을 모니터링하는 데 유용한 기준임을 시사한다.

주제어 : 암호화폐 시장, 비트코인, 통화정책, 시장 시가총액, 융합금융

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Ⅰ. Introduction

Digital-asset markets provide an important setting for data-driven convergence-finance analysis. Cryptocurrency market valuation is not determined only by blockchain-specific factors, but also by macro-financial conditions, investor attention, liquidity, and speculative demand. This feature makes cryptocurrency markets a useful empirical field in which digital-asset data, monetary-policy information, and statistical financial modelling can be integrated into a single analytical framework. In this context, the main issue is not only whether cryptocurrency prices fluctuate after policy events, but whether major policy-rate movements are followed by measurable changes in the valuation benchmark of the digital-asset market.

Bitcoin was initially introduced as a decentralised monetary technology, but it is now widely analysed as a speculative financial asset [1]. Previous asset-pricing research shows that cryptocurrency returns are affected by market-specific risk factors, momentum, investor attention, and adoption dynamics rather than by the behaviour of conventional currencies, commodities, or equities alone [2]. As cryptocurrency markets have become more connected to global financial conditions, it has become necessary to examine them not only as technological assets, but also as data-rich financial markets exposed to common macro-financial shocks.

U.S. monetary policy provides a direct empirical setting for this study. Monetary-policy decisions can affect financial asset prices through the current policy-rate target, expectations about future policy, discount rates, liquidity conditions, funding constraints, and risk appetite [3]. Standard event-study reasoning suggests that financial prices may adjust when new policy information becomes available [4]. However, It may not be limited to the immediate event window for the relevant adjustment in cryptocurrency markets. Because digital assets are highly sensitive to liquidity, speculative demand, and the opportunity cost of holding non-yielding assets, policy-rate changes may be followed by medium-run revaluation as well as short-run volatility.

A growing literature shows that cryptocurrency markets respond to monetary-policy news and macro-financial conditions. Prior studies document heterogeneous cryptocurrency reactions to Federal Open Market Committee announcements [5], Bitcoin responses to monetary-policy shocks [6], policy effects on Bitcoin valuation and its ecosystem [7], and state-dependent cryptocurrency price and volatility responses to U.S. monetary-policy shocks [8]. These studies establish that cryptocurrency markets are not fully isolated from monetary-policy transmission. However, less attention has been given to how policy-rate movements are reflected in medium-run cryptocurrency market capitalisation, especially when market-wide capitalisation is compared with Bitcoin-specific evidence.

This study addresses this issue by developing a data-driven convergence-finance framework for measuring cryptocurrency market revaluation after major U.S. monetary-policy movements. The framework combines three types of data and methods. First, it uses digital-asset market data, measured by total cryptocurrency market capitalisation (TMC) and Bitcoin market capitalisation (BTC). TMC is interpreted as a market-wide digital-asset valuation benchmark, while BTC is used as the dominant single-asset benchmark. Secondly, it uses U.S. policy-rate data to identify policy-shock months and to classify them into easing and tightening episodes. Thirdly, it applies econometric methods, including event-window statistics, local projections, and ridge-regularised distributed-lag diagnostics, to distinguish short-run volatility disturbance from medium-run valuation adjustment.

The empirical design separates three layers of post-shock response. The first layer is short-run volatility disturbance, measured by cumulative abnormal volatility. This measure examines whether volatility

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increases or decreases relative to its pre-event benchmark after policy-rate changes. The second layer is medium-run valuation displacement, measured by comparing post-event valuation levels with the pre-event benchmark. This measure is central to the study because it tests whether cryptocurrency market capitalisation remains systematically revalued after the immediate policy-event period. The third layer is cumulative horizon-specific adjustment, estimated using local projections and supported by ridge-regularised distributed-lag diagnostics. Local projections estimate cumulative valuation responses at fixed horizons [9], while ridge regularisation stabilises distributed-lag estimates in a finite monthly sample [10]. Heteroskedasticity-and-autocorrelation-consistent inference is used to account for serial dependence in cumulative responses [11].

This make three contributions to the literature. First, we propose a convergence-finance approach combining digital-asset data, monetary-policy indicators, and statistical financial modelling, which is well-suited for next-generation financial analysis because cryptocurrency valuations depend on both digital-market structures and macro-financial conditions. Second, we distinguish short-run volatility disturbances from medium-run valuation adjustments, as temporary volatility responses differ fundamentally from persistent valuation shifts that redefine market-wide digital-asset benchmarks. Third, we compare TMC and BTC to determine whether market-wide cryptocurrency capitalization reveals insights that a Bitcoin-only analysis obscures.

The empirical results support the usefulness of this framework. The main post-shock signal appears in valuation levels rather than in volatility measures. Easing episodes are followed by positive medium-run revaluation in both TMC and BTC, while tightening responses are weaker and less precisely estimated. Local-projection estimates show that the positive easing response becomes visible at 12- and 18-month horizons rather than immediately after the shock. The comparison between TMC and BTC suggests that market-wide capitalisation provides a useful benchmark for monitoring aggregate digital-asset adjustment beyond Bitcoin-specific movements.

The remainder of this paper is organised as follows. Section 2 reviews related studies and develops the empirical hypotheses. Section 3 explains the data, policy-shock classification, and methodology. Section 4 reports the empirical results on valuation displacement, local-projection responses, volatility disturbance, and distributed-lag diagnostics. Section 5 discusses the implications for data-driven convergence-finance analysis and digital-asset market monitoring. Section 6 concludes.

II. Related Literature

2.1 Cryptocurrency markets as convergence-finance data

Cryptocurrency markets are increasingly examined not only as blockchain-based technological systems, but also as financial markets that generate high-frequency and high-dimensional digital-asset data. Bitcoin was originally proposed as a decentralised monetary technology, but empirical studies show that it is also traded and held as a speculative financial asset [1]. Cryptocurrency returns are influenced by market-specific risk factors, adoption dynamics, momentum, investor attention, and crypto-specific premia rather than by the behaviour of traditional currencies, commodities, or equities alone [2]. These findings imply that cryptocurrency markets should be analysed as digital-asset financial markets in which technological characteristics and macro-financial conditions interact.

Recent articles in the Journal of Next-generation Convergence Technology Association have also examined cryptocurrency investment intention and Bitcoin-related

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technological infrastructure, confirming that cryptocurrency has been treated as a relevant object of next-generation convergence-technology research [12,13].

This perspective is important for convergence-finance research. Cryptocurrency valuation reflects both digital-market characteristics and broader financial conditions. As trading platforms, institutional access, stablecoin liquidity, and portfolio participation have expanded, cryptocurrency market capitalisation has become increasingly relevant for empirical financial analysis. Therefore, the central question is not whether cryptocurrency markets are fully separated from traditional finance, but how digital-asset valuation adjusts when common macro-financial conditions change.

2.2 Monetary-policy shocks and digital-asset valuation

Monetary policy is a major source of macro-financial information. U.S. monetary-policy decisions can affect financial asset prices through the current policy-rate target, expectations about the future policy path, discount rates, liquidity conditions, funding constraints, and risk appetite [3]. Event-study methods provide a standard approach for evaluating whether identified policy events generate abnormal financial-market responses [4]. In cryptocurrency markets, however, the relevant response may extend beyond the immediate announcement window because digital assets are highly sensitive to liquidity, speculative demand, and the opportunity cost of holding non-yielding assets.

Recent studies show that cryptocurrency markets respond to monetary-policy news. Prior research documents heterogeneous cryptocurrency reactions to Federal Open Market Committee announcements [5], Bitcoin price responses to monetary-policy shocks [6], the influence of discretionary monetary policy on Bitcoin valuation and its ecosystem [7], and state-dependent price and volatility responses to U.S. monetary-policy shocks [8]. These studies establish that cryptocurrency markets are not fully insulated from monetary-policy conditions. However, the existing literature has focused mainly on returns, volatility, or short-window announcement effects. Less attention has been paid to whether major policy-rate movements are followed by medium-run changes in cryptocurrency market capitalisation.

2.3 Persistent revaluation and market-wide measurement

The distinction between short-run volatility disturbance and persistent valuation adjustment is central to this study. A volatility response measures whether market fluctuations increase or decrease relative to a pre-event benchmark. A valuation-level response measures the market remains systematically revalued after the immediate event period. These two responses are not identical. Volatility may normalise while market capitalisation remains displaced, and a medium-run valuation shift may occur even when short-run volatility responses are statistically weak.

This distinction is valid for data-driven convergence-finance analysis. If the empirical focus is limited to short-run returns or volatility, the adjustment of cryptocurrency markets to monetary-policy movements may appear temporary. In contrast, market-capitalisation measures can reveal whether the digital-asset valuation benchmark changes over a longer horizon. Therefore, this study treats persistent revaluation as a separate empirical object and evaluates it using post-shock valuation displacement and horizon-specific cumulative responses.

A further issue concerns the appropriate measurement benchmark. Much of the cryptocurrency literature uses Bitcoin as the representative asset because of its size, liquidity, and long trading history. This approach is useful, but it may not fully capture common adjustment across the cryptocurrency market. Bitcoin contains both a market-wide digital-asset component

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and a Bitcoin-specific component. Total cryptocurrency market capitalisation (TMC) provides a broader benchmark because it captures aggregate repricing across digital assets. So this study compares TMC with Bitcoin market capitalisation (BTC) to distinguish market-wide digital-asset adjustment from Bitcoin-specific movement.

Based on this literature, the present study examines whether major U.S. policy-rate movements are associated with persistent cryptocurrency market revaluation, whether easing and tightening episodes generate asymmetric responses, and whether TMC provides a useful market-wide benchmark beyond BTC.

III. Data and Methodology

3.1 Data and variables

This study uses a monthly dataset that combines digital-asset market data and macro-financial policy data. The two main cryptocurrency variables are total cryptocurrency market capitalisation (TMC) and Bitcoin market capitalisation (BTC). TMC is used as a market-wide digital-asset valuation benchmark, while BTC is used as the dominant single-asset benchmark. This distinction allows the study to examine whether post-policy adjustment is a common market-wide phenomenon or mainly a Bitcoin-specific movement.

The policy variable is the U.S. policy-rate series (USIR). Weekly source observations for TMC, BTC, and USIR are aligned to monthly frequency using the last available observation in each calendar month. The monthly frequency is appropriate for this study because the objective is not to estimate intraday responses to policy announcements, but to examine whether major policy-rate movements are followed by medium-run changes in cryptocurrency market valuation.

Table 1. Variables and empirical design

Type	Variable	Definition and role
Market valuation	TMC	Log total cryptocurrency market capitalisation; market-wide digital-asset valuation benchmark
Market valuation	BTC	Log Bitcoin market capitalisation; dominant single-asset benchmark
Policy variable	USIR	U.S. policy-rate series; used for policy-shock classification
Volatility proxy	$(\mid r \mid)$	Absolute monthly log return; amplitude-type volatility proxy
Volatility proxy	(r^2)	Squared monthly log return; variance-type volatility proxy
Event measure	CAV	Cumulative abnormal volatility; short-run volatility disturbance
Event measure	DC	Post-shock level displacement; medium-run valuation displacement
Lag diagnostic	$(K(\tau))$	Ridge-regularised distributed-lag coefficient, $(\tau = 0, ..., 18)$
Lag diagnostic	$(Cum(H))$	Cumulative distributed-lag response, $(H = 0, ..., 18)$

Notes: TMC denotes total cryptocurrency market capitalisation, BTC denotes Bitcoin market capitalisation, and USIR denotes the U.S. policy-rate series. TMC and BTC are expressed as log market-capitalisation levels. Weekly source data are aligned to monthly frequency using month-end observations. Monthly log returns are defined from the log valuation series. The volatility proxies are $\mid r \mid$, the absolute monthly log return, and r^2, the squared monthly log return. Policy-shock months are classified by the sign of monthly changes in USIR, with positive changes treated as tightening episodes and negative changes treated as easing episodes.

The valuation variables are expressed in logarithms. Let $(y_{a,t})$ indicate the log market capitalisation of asset (a), where (a) is either TMC or BTC. Monthly log returns are defined as $r_{a,t} = y_{a,t} - y_{a,t-1}$.

Two volatility proxies are constructed from monthly

log returns. The first is the absolute return, ($|r_{a,t}|$), which captures the amplitude of monthly market movement. The second is the squared return, ($r_{a,t}^2$), which captures variance-type disturbance. These volatility proxies are used to examine whether policy-rate movements generate short-run volatility disturbance and whether such disturbance differs from medium-run valuation adjustment.

Policy-shock months are identified from non-zero monthly changes in USIR. In the present sample, this corresponds to policy-rate movements of at least 25 basis points in absolute value. Positive monthly changes are classified as tightening episodes, while negative monthly changes are classified as easing episodes. The final event set contains 27 policy-shock months, consisting of 20 tightening episodes and 7 easing episodes. This classification is intentionally simple and transparent. The full list of policy-rate events is reported in Appendix Table A1. It should be interpreted as a monthly policy-rate movement indicator, not as a high-frequency monetary-policy surprise measure.

3.2 Event-window measures

The first empirical step uses an event-window design to distinguish short-run volatility disturbance from medium-run valuation adjustment. For each policy-shock month (i), the pre-event benchmark is defined over months (-12) to (-3) relative to the event month. Months (-2) and (-1) are excluded as a buffer to reduce contamination from immediate pre-event positioning.

Post-event responses are evaluated over fixed horizons after the policy-shock month.

Short-run volatility disturbance is measured by cumulative abnormal volatility (CAV). For event (i), asset (a), volatility proxy (v), and horizon (H), CAV is defined as $CAV_{i,a,v}(H) = \sum_{h=0}^{H} (v_{i+h,a} - \overline{v_{i,a}^{pre}})$,

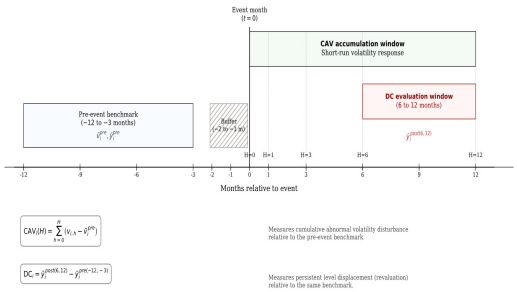


Fig. 1. Event-window design for measuring post-shock responses.

Notes: The pre-event benchmark is defined over months -12 to -3, followed by a buffer over months -2 to -1. The event month is t = 0. CAV measures post-event volatility accumulation from month 0 to horizon H, while DC measures medium-run valuation displacement over months 6 to 12.

where ($v_{i+h,a}$) denotes the volatility proxy after the event and ($\overline{v_{i,a}^{pre}}$) means the average value of the same volatility proxy during the pre-event benchmark window. A positive CAV indicates above-benchmark volatility accumulation after the policy movement, while a negative CAV indicates below-benchmark volatility.

Medium-run valuation adjustment is measured by a post-shock displacement statistic. For event (i) and asset (a), the displacement measure is defined as $DC_{i,a} = \overline{y^{post}}i,a - \overline{y^{pre}}i,a$. In the equation, ($\overline{y^{pre}}i,a$) is the average log valuation level during the pre-event benchmark window and ($\overline{y^{post}}i,a$) is the average log valuation level during months 6 to 12 after the event. A positive value of ($DC_{i,a}$) indicates that cryptocurrency market capitalisation remains above its pre-event benchmark in the medium run. A negative value indicates below-benchmark valuation. This measure is central to the study because it directly tests whether major policy-rate movements are followed by persistent changes in digital-asset market valuation.

3.3 Local-projection model

The second empirical step estimates cumulative

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valuation responses using local projections. Local projections are useful because they estimate horizon-pecific responses directly without imposing a full dynamic system [9]. For asset (a), the dependent variable is the cumulative log-level change from month (t) to month (t+H), multiplied by 100 for interpretability: $100(y_{a,t+H}-y_{a,t})$.

The baseline local-projection specification is

$$100(y_{a,t+H}-y_{a,t})=\alpha_{a,H}+\beta^{E}_{a,H}Ease_t.$$
$$+\beta^{T}_{a,H}Tight_t+\Gamma_{a,H}X_{t-1}+\varepsilon_{a,t,H}.$$

The ($Ease_t$) means easing policy-rate movements and ($Tight_t$) is tightening policy-rate movements. The coefficients ($\beta^{E}_{a,H}$) and ($\beta^{T}_{a,H}$) measure cumulative valuation responses at horizon (H). The vector (X_{t-1}) includes lagged monthly log returns and lagged policy-shock variables. The main horizons are 6, 12, and 18 months. Standard errors are computed using heteroskedasticity-and-autocorrelation-consistent Newey - West inference [11], because cumulative horizon outcomes may contain serial dependence.

This model examines if post-shock revaluations remain apparent when valuation changes are estimated at fixed horizons. Also it allows easing and tightening episodes to be examined separately, rather than imposing a symmetric policy-response structure.

3.4 Ridge-regularised distributed-lag diagnostics

The third empirical step uses ridge-regularised distributed-lag diagnostics to examine whether the valuation response is concentrated immediately after the policy movement or gradually accumulates over time. This step is diagnostic rather than the main source of identification.

Separate distributed-lag profiles are estimated for easing and tightening episodes. Let ($K^{E}_{a}(\tau)$) and ($K^{T}_{a}(\tau)$) denote the lag coefficients for easing and tightening policy movements at lag (τ). The cumulative distributed-lag response up to horizon (H) is defined as $CR^{E}_{a}(H)=\sum_{\tau=0}^{H}K^{E}_{a}(\tau)$, $CR^{T}_{a}(H)=\sum_{\tau=0}^{H}K^{T}_{a}(\tau)$.

The lag window is set from 0 to 18 months. Ridge regularisation is used to reduce instability in distributed-lag coefficients and to produce smoother lag profiles in a finite monthly sample [10]. The main diagnostic quantities are the cumulative response, the tail area over later lags, the peak lag, and the half-life of the response. Uncertainty intervals are evaluated using a moving-block bootstrap, which is appropriate when monthly financial observations may exhibit serial dependence [14].

3.5 Empirical strategy

The empirical strategy links each method to a specific part of the convergence-finance question. CAV measures short-run volatility disturbance. The displacement statistic measures whether the digital-sset valuation benchmark remains shifted in the medium run. Local projections estimate horizon-specific cumulative valuation responses. Ridge-regularised distributed-lag diagnostics provide additional timing evidence, but they are not used as the primary basis for inference.

This structure allows the study to evaluate whether major U.S. policy-rate movements are followed mainly by temporary volatility disturbance or by persistent medium-run cryptocurrency market revaluation. It also allows the comparison between TMC and BTC to determine whether market-wide capitalisation provides information that is not fully captured by a Bitcoin-specific benchmark.

Ⅳ. Empirical Results

4.1 Post-shock valuation displacement

This section reports the empirical results on

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cryptocurrency market adjustment after major U.S. policy-rate movements. The first question is whether policy-rate changes are followed mainly by short-run volatility disturbance or by medium-run valuation adjustment. To answer this question, the analysis begins with the post-shock displacement statistic, (DC), which compares the medium-run post-event valuation level with the pre-event benchmark.

Table 2. Post-shock displacement by asset and shock type

Group	N	Level DC(t-stat)	Volatility DC(t-stat)
Easing (TMC)	4	0.699(3.654)	r^2: -0.003 (-0.197)
Easing (BTC)	4	0.779(5.113)	r^2: -0.007 (-0.372)
Tightening (TMC)	20	0.506(1.713)	r^2: -0.018 (-0.919)
Tightening (BTC)	20	0.500(2.106)	r^2: -0.009 (-0.690)

Notes: DC denotes post-shock displacement relative to the pre-event benchmark. Values in parentheses under Level DC are t-statistics. For the level metric, DC is computed as the difference between the average log market capitalisation in the post-event window and the pre-event benchmark mean. For volatility metrics, DC is computed using $|r|$, the absolute monthly log return, and r^2, the squared monthly log return. The pre-event benchmark is defined over months -12 to -3, and the post-event window is defined over months 6 to 12.

Table 2 reports post-shock displacement statistics for valuation levels and volatility proxies. The central result is that the strongest post-shock signal appears in valuation levels rather than in short-run volatility measures. After easing episodes, the mean level displacement is 0.6990 for TMC and 0.7791 for BTC. The corresponding t-statistics are 3.6540 and 5.1127, respectively. These results indicate that easing episodes are followed by positive medium-run revaluation in both total cryptocurrency market capitalisation and Bitcoin market capitalisation.

Tightening episodes show weaker and less stable evidence. The mean level displacement is 0.5055 for TMC and 0.4998 for BTC, with t-statistics of 1.7128 and 2.1065, respectively. However, this result should be interpreted cautiously because later local-projection estimates do not show statistically significant tightening responses. Therefore, the tightening evidence is better understood as weaker and less robust across specifications, rather than as a clear positive revaluation effect.

The volatility-based displacement measures are generally small and statistically weak. For both TMC and BTC, the estimates based on (|r|) and (r^2) are close to zero or negative, and their t-statistics are generally far from conventional significance levels. This contrast suggests that the main empirical signal is not short-run volatility disturbance. Instead, the evidence is more consistent with medium-run valuation adjustment in cryptocurrency market capitalisation.

4.2 Local-projection cumulative responses

The second step examines whether the valuation-isplacement result remains visible when cumulative valuation changes are estimated directly at fixed horizons. Fig. 2 and Table 3 report local-projection estimates at 6-, 12-, and 18-month horizons. The dependent variable is the cumulative log-level change from month (t) to month (t+H), multiplied by 100 for interpretability.

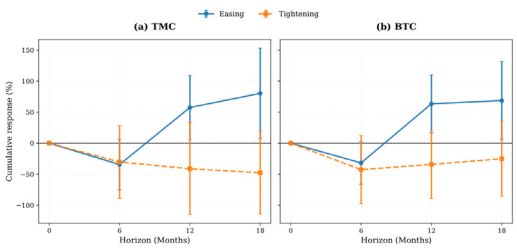


Fig. 2. Local-projection cumulative valuation responses.

Notes: The figure reports cumulative valuation responses for TMC and BTC at 6-, 12-, and 18-month horizons after U.S. policy-rate movements. Error bars denote approximate 95% Newey-West confidence intervals. Easing and tightening episodes are estimated separately.

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At the 6-month horizon, the easing response is still negative for both TMC and BTC. The TMC estimate is -34.3745, while the BTC estimate is -31.7823. These results indicate that the positive valuation response to easing does not appear immediately after the policy movement.

At longer horizons, the response changes direction and becomes statistically significant. For TMC, the easing response is 57.4190 at the 12-month horizon and 80.2594 at the 18-month horizon. For BTC, the corresponding estimates are 63.4333 and 68.6692. The confidence intervals exclude zero at these medium-un horizons. This pattern indicates delayed cumulative revaluation rather than immediate repricing.

Table 3. Local-projection cumulative responses

Group	H = 6	H = 12	H = 18
Easing (TMC)	-34.375* (-1.648)	57.419** (2.193)	80.259** (2.152)
Easing (BTC)	-31.782* (-1.825)	63.433*** (2.659)	68.669** (2.138)
Tightening (TMC)	-30.307 (-1.013)	-41.200 (-1.084)	-47.753 (-1.408)
Tightening (BTC)	-42.594 (-1.518)	-34.200 (-1.213)	-25.128 (-0.816)

Notes: The table reports local-projection estimates of cumulative valuation responses. Values in parentheses are Newey-West t-statistics. The dependent variable is the cumulative log-level change from month t to month $t+H$, multiplied by 100. Easing and tightening episodes are defined from negative and positive monthly changes in USIR, respectively. The specification includes contemporaneous policy-rate movements, six lags of monthly log returns, and six lags of policy-shock variables. The number of observations is 113 for $H=6$, 107 for $H=12$, and 101 for $H=18$. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Tightening responses are negative at all reported horizons, but they are not statistically significant. For TMC, the tightening estimates are -30.3071 at 6 months, -41.1995 at 12 months, and -47.7528 at 18 months. For BTC, the corresponding estimates are -42.5938, -34.1998, and -25.1279. In all cases, the confidence intervals include zero. The local-projection evidence therefore supports the interpretation that easing and tightening episodes are not symmetric. Easing is followed by clearer medium-run revaluation, whereas tightening effects are weaker and less precisely estimated.

4.3 Volatility disturbance and distributed-lag diagnostics

The cumulative abnormal volatility diagnostics provide a contrast with the valuation-level evidence. The diagnostics indicate that volatility responses are generally modest and imprecisely estimated across shock types, assets, and horizons. The exception is a limited number of BTC easing cells, where volatility declines at some medium horizons. However, this pattern is narrower than the valuation-displacement evidence and does not provide the main basis for the study's interpretation.

The ridge-regularised distributed-lag diagnostics provide additional information on the timing of adjustment. Selected distributed-lag diagnostics are reported in Appendix Table A2. The diagnostics indicate that easing profiles are directionally more persistent than tightening profiles. The estimated half-life is 11 months for TMC and 9 months for BTC, while the peak easing lag occurs at month 17 for TMC and month 9 for BTC. However, the bootstrap confidence intervals are wide. Therefore, the distributed-lag results should be treated as conservative diagnostics rather than standalone evidence of persistence.

4.4 Summary of empirical findings

The empirical results provide three main findings. First, major U.S. policy-rate movements are followed by stronger signals in cryptocurrency valuation levels than in short-run volatility measures. This supports the view that medium-run revaluation is a distinct

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empirical object in digital-asset markets. Secondly, easing and tightening episodes are not mirror-image events. Easing is followed by positive and statistically identifiable revaluation at 12- and 18-month horizons, while tightening responses are weaker and statistically insignificant in the local-projection estimates. Thirdly, the comparison between TMC and BTC suggests that market-wide cryptocurrency capitalisation provides useful information beyond Bitcoin-specific movements.

Taken together, the results support the usefulness of a data-driven convergence-finance framework. The framework, which combines digital-asset market data, monetary-policy information, and econometric response analysis, distinguishes temporary volatility disturbances from persistent valuation adjustments in cryptocurrency markets.

Ⅴ. Discussion

5.1 Medium-run revaluation as a convergence-finance signal

The empirical results indicate that cryptocurrency-market adjustment after major U.S. policy-rate movements is more clearly observed in valuation levels than in short-run volatility measures. The post-shock displacement estimates show positive medium-run revaluation after easing episodes, while the local-projection estimates indicate that this response becomes visible mainly at 12- and 18-month horizons. In contrast, the volatility-based measures are generally small and statistically weak. This pattern suggests that the main empirical signal is not temporary volatility disturbance, but medium-run adjustment in digital-asset market capitalisation.

This finding is important for data-driven convergence-finance analysis. Cryptocurrency markets combine digital-asset trading data, speculative demand, platform-based liquidity, and exposure to macro-financial conditions. If the study focuses only on short-run returns or volatility, the adjustment of cryptocurrency markets to policy-rate movements may appear weak or temporary. But valuation-level measures show that the market capitalisation benchmark itself remains displaced after the immediate event period. So the present results suggest that medium-run revaluation should be treated as a separate empirical object in analysing digital-asset markets.

The contribution of this study is not to claim that policy-rate movements mechanically cause cryptocurrency valuation changes. The shock classification is based on observed monthly policy-rate changes, not high-frequency monetary-policy surprises. Therefore, the results should be interpreted as evidence that major policy-rate movements are followed by measurable medium-run changes in cryptocurrency market capitalisation. This interpretation is consistent with the purpose of the study: to construct a transparent convergence-finance framework that integrates digital-asset data, monetary-policy information, and econometric response analysis.

5.2 Asymmetric responses to easing and tightening episodes

The results also suggest that easing and tightening episodes should not be treated as symmetric policy events. Easing episodes are followed by positive and statistically identifiable medium-run revaluation in both total cryptocurrency market capitalisation and Bitcoin market capitalisation. By contrast, tightening responses are weaker and statistically insignificant in the local-projection estimates. The asymmetry is not simply a matter of opposite signs. It concerns the magnitude, timing, and precision of post-shock valuation adjustment.

This pattern is consistent with macro-financial mechanisms through which accommodative policy may support speculative asset valuation. Easing can

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reduce discount-rate pressure, improve risk appetite, and relax liquidity or funding conditions [15]. Liquidity conditions may also interact with funding constraints, producing amplification effects in financial markets [16]. In addition, monetary-policy announcements may contain information and expectation effects that extend beyond contemporaneous changes in the policy-rate target [17]. These mechanisms suggest why easing and tightening may differ in their cumulative effects on cryptocurrency markets.

For digital-asset markets, this asymmetry is particularly relevant. Cryptocurrencies are non-yielding and highly valuation-sensitive assets. Their market capitalisation can be affected by liquidity expectations, speculative participation, and changes in the opportunity cost of holding risky assets. Therefore, an easing episode may support delayed revaluation even when the immediate response is weak. Tightening may operate through opposite channels, but the adjustment may be less stable because market participants adjust leverage, liquidity positions, and portfolio exposure unevenly. The results support this asymmetric interpretation.

5.3 Market-wide capitalisation as a digital-asset benchmark

A further implication concerns the measurement benchmark. Bitcoin keeps the dominant single-asset reference point in cryptocurrency research because of its size, liquidity, and long trading history. However, Bitcoin does not fully represent the cryptocurrency market as a whole. Bitcoin contains both a common digital-asset component and a Bitcoin-specific component. Therefore, a Bitcoin-only analysis may obscure part of the common market-wide adjustment to macro-financial conditions.

Total cryptocurrency market capitalisation provides a broader benchmark for convergence-finance analysis. It captures aggregate repricing across the digital-asset market and is more closely aligned with the question of market-wide valuation adjustment. In the present results, both TMC and BTC show positive medium-run revaluation after easing episodes. but TMC is conceptually better suited to measuring common adjustment across the cryptocurrency market. The distributed-lag diagnostics also provide directional evidence that the market-wide response may have a gradual adjustment profile, although the wide bootstrap intervals require cautious interpretation.

The conclusion is not that Bitcoin is uninformative. Rather, Bitcoin should be interpreted as a dominant single-asset benchmark, while TMC should be interpreted as a market-wide digital-asset valuation benchmark. Using both measures allows the research to distinguish Bitcoin-specific movement from broader cryptocurrency-market adjustment. This distinction strengthens the convergence-finance framing of the study because it connects asset-level and market-wide digital-finance data within the same empirical framework.

5.4 Implications and limitations

The results have practical implications for monitoring digital-asset markets. First, short-run volatility alone is an incomplete indicator of post-policy cryptocurrency adjustment. The stronger signal in this study appears in medium-run valuation levels. This implies that market-monitoring frameworks should include valuation displacement measures as well as volatility indicators. Secondly, easing and tightening episodes should be modelled separately. A symmetric policy-shock specification may miss differences in timing, strength, and precision between easing and tightening responses. Thirdly, total cryptocurrency market capitalisation should be used alongside Bitcoin when the objective is to evaluate market-wide digital-asset adjustment.

Several limitations should be noted. First, the study uses observed monthly changes in the U.S. policy-rate series. It does not identify high-frequency monetary-

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policy surprises. Therefore, the results should not be interpreted as intraday causal estimates of unexpected policy news. Second, the number of complete easing windows is small, especially at longer post-event horizons. This seems that the easing results, although informative, should be interpreted cautiously. Thirdly, monthly alignment may smooth within-month reactions around Federal Open Market Committee announcements. Finally, the analysis is reduced-form and does not directly identify the transmission channels through which stablecoin supply, leverage, funding rates, institutional participation, exchange liquidity, or global dollar liquidity affect cryptocurrency valuation.

Future research can extend the present framework in several directions. One extension is to incorporate market-state variables such as stablecoin supply, exchange liquidity, derivatives funding rates, and leverage indicators. Another extension is to examine whether the revaluation response differs across periods of market stress, institutional adoption, or global liquidity tightening. A further extension is to compare total market capitalisation with other digital-asset segments, such as altcoin capitalisation or decentralised-finance tokens. These extensions would clarify the channels through which macro-financial policy conditions interact with digital-asset market valuation.

Ⅵ. Conclusion

This study proposed a data-driven convergence-finance framework for analysing cryptocurrency market revaluation after major U.S. policy-rate movements. By combining digital-asset market data, monetary-policy information, and econometric response analysis, the study examined if cryptocurrency markets adjust mainly through short-run volatility disturbance or through medium-run valuation displacement.

The empirical analysis used total cryptocurrency market capitalisation (TMC) and Bitcoin market capitalisation (BTC) as the main valuation measures. TMC was used as a market-wide digital-asset valuation benchmark. while, BTC was used as the dominant single-asset benchmark. Policy-rate movements were classified into easing and tightening episodes, and post-shock responses were evaluated using event-window statistics, local projections, and ridge-regularised distributed-lag diagnostics.

The main finding is that the stronger empirical signal appears in valuation levels rather than in short-run volatility measures. Easing episodes are followed by positive medium-run revaluation in both TMC and BTC, while tightening responses are weaker and less precisely estimated. The local-projection estimates show that the positive easing response becomes visible at 12- and 18-month horizons, rather than immediately after the policy movement. This suggests that cryptocurrency market adjustment to monetary-policy conditions is not fully captured by short-window returns or volatility indicators.

The results also show that easing and tightening episodes should not be treated as mechanically symmetric events. The asymmetry concerns the magnitude, timing, and statistical precision of the valuation response. In addition, the comparison between TMC and BTC indicates that total cryptocurrency market capitalisation provides a useful market-wide benchmark for observing aggregate digital-asset adjustment, while Bitcoin remains an important but more asset-specific reference point.

The contribution of this study is to show how digital-asset data, macro-financial policy information, and statistical financial methods can be integrated into a convergence-finance framework. The novelty is not simply to show that cryptocurrency markets respond to policy-rate movements, but to distinguish temporary volatility disturbance from medium-run valuation adjustment and to compare market-wide

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capitalisation with Bitcoin-level evidence.

Several limitations remain. The policy-shock classification is based on observed monthly policy-rate changes rather than high-frequency monetary-policy surprises. The number of complete easing windows is limited, especially at longer horizons. Also the analysis is reduced-form and does not directly identify the channels through which stablecoin supply, leverage, funding rates, exchange liquidity, institutional participation, or global dollar liquidity affect cryptocurrency valuation. Future research can extend the proposed framework by incorporating these market-state variables and by examining whether digital-asset revaluation differs across liquidity regimes, market-stress periods, and cryptocurrency market segments.

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Appendix. Additional Event Classification and Robustness Results

Appendix Table A1. Policy-rate event classification

Event month	Type	($\Delta USIR$)
2015-12	Tightening	0.25
2016-12	Tightening	0.25
2017-03	Tightening	0.25
2017-06	Tightening	0.25
2017-12	Tightening	0.25
2018-03	Tightening	0.25
2018-06	Tightening	0.25
2018-09	Tightening	0.25
2018-12	Tightening	0.25
2019-07	Easing	-0.25
2019-09	Easing	-0.25
2019-10	Easing	-0.25
2020-03	Easing	-1.50
2022-03	Tightening	0.25
2022-05	Tightening	0.50
2022-06	Tightening	0.75
2022-07	Tightening	0.75
2022-09	Tightening	0.75
2022-10	Tightening	0.75
2022-12	Tightening	0.50
2023-01	Tightening	0.25
2023-03	Tightening	0.25
2023-05	Tightening	0.25
2023-07	Tightening	0.25
2024-09	Easing	-0.50
2024-11	Easing	-0.25
2024-12	Easing	-0.25

Notes: $\Delta USIR$ indicates the monthly change in the U.S. policy-rate series, measured in percentage points. Positive changes are classified as tightening episodes, while negative changes are classified as easing episodes. The final event set contains 27 policy-rate movements: 20 tightening episodes and 7 easing episodes.

Appendix Table A2. Distributed-lag diagnostic summary

Metric	TMC	BTC
Tail area	0.0026	0.0019
Cumulative easing response	0.0022	0.0017
Cumulative tightening response	-0.0024	-0.0007
Half-life, months	11	9
Peak lag, months	17	9

Notes: The table reports selected ridge-regularised distributed-lag diagnostics. The estimates are interpreted as conservative timing diagnostics rather than as standalone evidence of persistence. Bootstrap confidence intervals are wide and are therefore not used as the primary basis for inference.